

Results and Interpretation

Implementation outcomes, API behaviour, and performance interpretation

Successfully implemented an event-driven data pipeline that processes uploaded CSV datasets automatically, stores graph data in Neo4j, and enables conversational querying through a chatbot.

Implementation Results

- CSV upload completed successfully.
- Dataset validation performed before ingestion.
- Data cleaning completed automatically.
- Kafka streamed each CSV row independently.
- Neo4j stored graph nodes and relationships.
- Chatbot returned responses from stored data.

CSV Processing Results

- File format validated.
- Duplicate records removed.
- Missing values handled.
- Data standardized.
- Clean dataset generated.

Kafka Results

- Producer published each row as a message.
- Consumer received messages successfully.
- Reliable asynchronous communication achieved.
- Improved fault tolerance.

Neo4j Results

- Nodes created successfully.
- Relationships established correctly.
- Duplicate entries prevented using MERGE.
- Efficient graph traversal enabled.

Chatbot Results

- Accepted natural language questions.
- Converted questions into Cypher queries.
- Retrieved accurate information from Neo4j.
- Returned grounded responses.

API Results

Endpoint	Purpose
POST /ingest	Upload dataset
POST /chat	Process user questions
GET /status	Monitor ingestion progress
GET /health	Verify service availability

Performance Interpretation

- Faster CSV processing.
- Reliable message delivery.
- Modular architecture.
- Easy system scalability.
- Efficient graph querying.

Advantages

- Dynamic CSV support.
- Event-driven processing.
- Graph-based storage.
- Natural language interaction.
- Modular implementation.

Current Limitations

- Supports CSV files only.
- Basic natural language interpretation.
- Single Neo4j instance.
- No authentication.
- Limited analytics.

Future Improvements

- Excel and JSON support.
- Authentication.
- Cloud deployment.
- AI-assisted query interpretation.
- Analytics dashboard.
- Multi-user support.